

Enthusiast COURSE

PHASE : (RBT-3)

TARGET : PRE-MEDICAL 2025

Test Type : MAJOR

Test Pattern : NEET (UG)

TEST DATE : 04-04-2025

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	1	3	4	3	1	4	3	3	1	2	1	1	1	1	4	4	3	1	3	3	2	3	3	1	4	2	1	3	1	3
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	4	4	2	1	4	1	4	2	2	1	1	4	1	2	2	2	1	3	2	1	2	4	1	1	2	3	1	3	1	1
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	3	2	1	1	1	2	4	1	3	2	3	1	2	2	2	2	2	4	2	3	4	4	1	4	3	3	4	2	2
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	2	4	1	2	4	2	2	3	4	3	1	2	2	4	2	4	1	4	2	3	2	1	2	3	4	4	4	3	1	2
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	1	3	4	2	4	2	4	3	4	2	2	4	3	2	2	1	3	1	4	2	3	2	3	2	4	2	1	3	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	1	3	2	2	4	1	2	3	3	2	4	2	3	4	1	3	3	4	1	1	1	4	1	3	1	4	3	3	1	1

HINT - SHEET

1. Ans (1)

$$\vec{r}_{cm} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2}$$

2. Ans (3)

com of cubes will be

$$\left(\frac{1}{2}, \frac{3}{2}, \frac{1}{2}\right), \left(\frac{3}{2}, \frac{1}{2}, \frac{1}{2}\right), \left(\frac{1}{2}, \frac{1}{2}, \frac{3}{2}\right), \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$$

$$x_{cm} = \frac{\frac{1}{2}m + \frac{3}{2}m + \frac{1}{2}m + \frac{1}{2}m}{m + m + m + m} = \frac{3}{4}$$

Similarly,

$$y_{cm} = \frac{3}{4}$$

$$z_{cm} = \frac{3}{4}$$

3. Ans (4)

$$h_n = e^{2n}h \Rightarrow h_2 = e^{2 \times 2}h = e^4h$$

$$\text{Given, } h_2 = \frac{h}{n} \Rightarrow \frac{h}{n} = e^4h \Rightarrow e = \left(\frac{1}{n}\right)^{1/4}$$

4. Ans (3)

$$y_{cm} = \frac{m_1 y_1 + m_2 y_2 + m_3 y_3 + m_4 y_4 + m_5 y_5}{m_1 + m_2 + m_3 + m_4 + m_5}$$

$$= \frac{m(0) + m(0) + m(b) + m(b) + m(2b)}{5m} = \frac{4b}{5}$$

6. Ans (4)

Let the mass of ring having density λ be M. Center of mass

$$\frac{M_1 x_1 + M_2 x_2}{M_1 + M_2} = \frac{-M \frac{2R}{\pi} + 3M \cdot \frac{2R}{\pi}}{4M} = \frac{R}{\pi}$$

7. Ans (3)

$$v^1 = v_{rel.} + 3 = 9$$

8. Ans (3)

When the block is moving over wedge, let their common horizontal velocity be v

By the momentum conservation

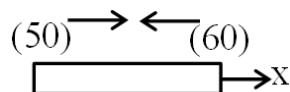
$$mv_0 = (m + M)v \quad \dots(i)$$

$$\frac{1}{2}mv_0^2 = \frac{1}{2}(m + M)v^2 + mgh \quad \dots(ii)$$

$$\frac{1}{2}mv_0^2 = \frac{1}{2}(m + M)\left(\frac{mv_0}{m + M}\right)^2 + mgh$$

$$v_0^2 \left(1 - \frac{m}{m + M}\right) = 2h \Rightarrow v_0 = \sqrt{\frac{2gh(m + M)}{M}}$$

10. Ans (2)



$$\Rightarrow 50(x + 2) + 60(x - 2) + 40x = 0$$

$$-20 + 150x = 0$$

$$x = \frac{2}{15}$$

12. Ans (1)

$$2mv_0 = mv_1 + mv_2$$

$$v_1 + v_2 = 2v_0 \quad \dots(i)$$

$$v_2 - v_1 = e2v_0 \quad \dots(ii)$$

From (i) + (ii)

$$2v_2 = 2v_0(1 + e)$$

$$v_2 = v_0(1 + e)$$

From (i) - (ii)

$$2v_1 = 2v_0(1 - e)$$

$$v_1 = v_0(1 - e)$$

$$\frac{v_1}{v_2} = \frac{1 - e}{1 + e}$$

13. Ans (1)

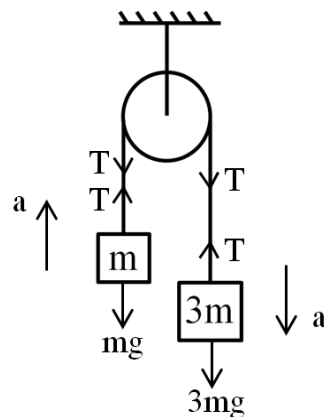
Let plank shifted by x then $\Delta x_{CM} = 0$, as there is no external force on the system.

$$m(L - x_{\text{plank}}) - Mx_{\text{plank}} = 0$$

$$X_{\text{plank}} = \frac{mL}{M + m}$$

14. Ans (1)

Telegram: @NEETxNOAH



$$3mg - T = 3ma$$

$$T - mg = ma$$

$$2mg = 4ma$$

$$a = \frac{2g}{4}$$

$$\therefore a = \frac{g}{2} \text{ m/s}^2$$

Now, acceleration of COM of the system,

$$\vec{a}_{\text{com}} = \frac{m_1\vec{a}_1 + m_2\vec{a}_2}{m_1 + m_2}$$

$$= \frac{3m \times \frac{g}{2} + m \times \left(-\frac{g}{2}\right)}{3m + m}$$

$$\Rightarrow \vec{a}_{\text{com}} = \left(\frac{3g}{2} - \frac{g}{2}\right) \times \frac{1}{4}$$

$$\Rightarrow \vec{a}_{\text{com}} = \frac{g}{4} \text{ m/s}^2$$

16. Ans (4)

$$\vec{I} = \Delta \vec{p} = m\Delta \vec{v}$$

$$\Rightarrow \Delta \vec{v} = \frac{\vec{I}}{m}$$

17. Ans (3)

$$I = \int F dt = \text{Area of F-t graph}$$

$$= \frac{1}{2} \times 2 \times 2 = 2 \text{ kg-m/s}$$

18. Ans (1)

Velocity of particles changes along line of impact during collision.

19. Ans (3)

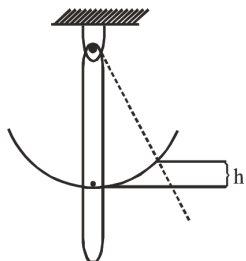
Velocity exchange in elastic collision

So $m_1 = m_2$

$$\frac{4}{3}\pi r^3 \rho_1 = \frac{4}{3}\pi \left(\frac{r}{2}\right)^3 \rho_2$$

$$\rho_2 = 8\rho_1$$

20. Ans (3)

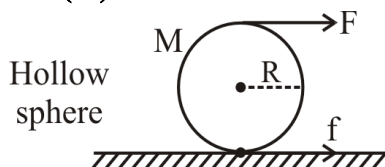


$$\frac{1}{2}I\omega^2 = mgh$$

$$\frac{1}{2} \frac{ml^2}{3} \omega^2 = mgh$$

$$h = \frac{1}{6} \frac{l^2 \omega^2}{g}$$

24. Ans (1)



$$F + f = Ma \quad \dots(1)$$

$$FR - fR = I\alpha = \frac{2}{3}MR^2 \frac{a}{R}$$

$$F - f = \frac{2}{3}Ma \quad \dots(2)$$

Add equation (1) & (2)

$$2F = \frac{5Ma}{3}$$

$$\therefore a = \frac{6F}{5m}$$

28. Ans (3)

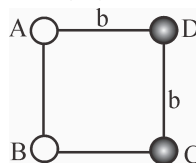
$$\theta = 5t^2 - 2t^3$$

$$w = \frac{d\theta}{dt} = 10t - 6t^2 \quad t = 3\text{sec}, w = 123 \text{ rad/s}, t = 6 \text{ sec}, w = 516 \text{ rad/s}$$

$$\alpha = \frac{dw}{dt} = 10 - 12t \quad t = 3\text{sec}, \alpha = 86 \text{ rad/s}^2, t = 6 \text{ sec}, \alpha = 176 \text{ rad/s}^2$$

30. Ans (3)

Telegram: @NEETxNOAH



Moment of inertia of a hollow sphere of radius R

about the diameter passing through D is

$$I_A = I_B = \frac{2}{3}MR^2 \quad \dots(i)$$

$$I_C = I_D = \frac{2}{5}MR^2 \quad \dots(ii)$$

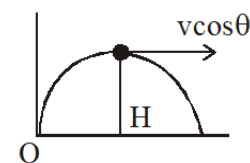
Moment of inertia of whole system about side

$$I_D = I_A + I_D + I_B + I_C$$

$$\frac{2}{3}MR^2 + \frac{2}{5}MR^2 + \left(Mb^2 + \frac{2}{3}MR^2\right) + \left(Mb^2 + \frac{2}{5}MR^2\right) = \frac{32}{15}MR^2 + 2Mb^2$$

32. Ans (4)

$$L = mv \cos \theta H$$



$$L = mv \cos \theta \frac{v^2 \sin^2 \theta}{2g}$$

$$L = \frac{mv^3 \sin^2 \theta \cos \theta}{2g}$$

34. Ans (1)

$$\vec{L} = \vec{r} \times \vec{P} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -1 \\ 3 & 4 & -2 \end{vmatrix}$$

$$\vec{L} = \hat{i}(-4+4) - \hat{j}(-2+3) + \hat{k}(4-6)$$

$$\vec{L} = 0\hat{i} - 1\hat{j} - 2\hat{k}$$

and the x axis is given by $\hat{i} + 0\hat{j} + 0\hat{k}$.

Dot product of these two vectors is zero so angular momentum is perpendicular to x axis.

35. Ans (4)

$$\text{as, } KE = \frac{L^2}{2I}$$

if $L = \text{constant}$,

$$\text{then } KE \propto \frac{1}{I}$$

as $I_A > I_B$

so $(KE)_A < (KE)_B$

36. Ans (1)

$$\therefore \vec{L}_P = \vec{L}_{cm} + \vec{r} \times \vec{p}_{cm} = I_{cm}\vec{\omega} + \vec{R} \times M\vec{v}_{cm};$$

here $v_{cm} = v_0$

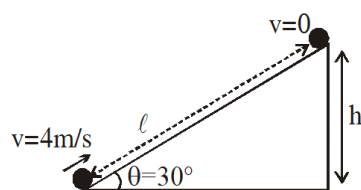
Since sphere is in pure rolling motion hence,

$$\omega = v_0/R$$

$$\Rightarrow \vec{L}_P = \left(\frac{2}{5}MR^2 \frac{v_0}{R} \right) (-\hat{k}) + Mv_0R(-\hat{k})$$

$$= \frac{7}{5} Mv_0R(-\hat{k})$$

37. Ans (4)



$$\frac{1}{2}mv^2 \left(1 + \frac{K^2}{R^2} \right) = mgh$$

$$\Rightarrow 8 \left(1 + \frac{1}{2} \right) = 10h$$

$$\Rightarrow h = 1.2 \text{ m}$$

$$\frac{h}{l} = \sin 30^\circ$$

$$\Rightarrow l = 2.4 \text{ m}$$

38. Ans (2)

Initially, angular momentum is, say, L .

Later, $L' = L + 200\% \text{ of } L$

$$\Rightarrow L' = 3L$$

Initial KE of rotating body

$$E_K = \frac{1}{2} \frac{L^2}{I}$$

$$\text{Later, } E'_k = \frac{1}{2} \frac{(L')^2}{I} = \frac{9}{2} \frac{L^2}{I}$$

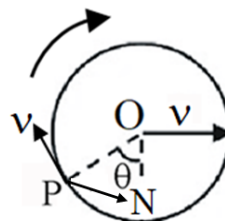
$$\Rightarrow E'_k = 9E_k$$

% increase in KE

$$= \frac{9E_k - E_k}{E_k} \times 100 = 800\%$$

41. Ans (1)

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Net speed of particle 'P' will be

$$V_R = \sqrt{V^2 + N^2 + 2V^2 \cos(180 - \theta)}$$

$$= \sqrt{2V^2(1 - \cos \theta)}$$

$$= \sqrt{2V^2 \left(1 - 1 + 2\sin^2 \frac{\theta}{2} \right)}$$

$$= 2V \sin \frac{\theta}{2}$$

42. Ans (4)

$$W = \frac{1}{2} mv^2 \left(1 + \frac{k^2}{R^2} \right)$$

$$= \frac{1}{2} (100) \left(\frac{1}{10} \right)^2 (1 + 1) = 1 \text{ J}$$

43. Ans (1)

$$I = \frac{2}{5} \left(m \left(\frac{R}{2} \right)^2 \right) + \left(\frac{2}{5} m \left(\frac{R}{2} \right)^2 + m(2R)^2 \right)$$

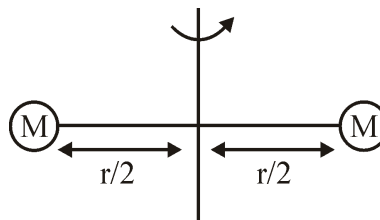
$$= \frac{2}{5} \frac{mR^2}{4} + \frac{2}{5} \frac{mR^2}{4} + 4mR^2$$

$$= mR^2 \left(\frac{1}{10} + \frac{1}{10} + 4 \right)$$

$$= mR^2 \left(\frac{1}{5} + 4 \right)$$

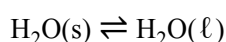
$$= \frac{21}{5} mR^2$$

44. Ans (2)



$$I = M_1 r_1^2 + M_2 r_2^2$$

49. Ans (2)



On pressure decrease, more ice will form.

55. **Ans (2)**

NCERT-XI, Pg. # 202

62. **Ans (3)**

$$[H^+] = 10^{-3} \text{ and } [H^+] = 10^{-5}$$

same volume = V.

$$[H^+]_{\text{net}} = \frac{10^{-3}V + 10^{-5}V}{2V} \approx \frac{10^{-3}}{2}$$

$$\text{pH} = 3 + \log 2 = 3.3$$

69. **Ans (1)****Explanation**

The pH of a solution is a measure of the hydrogen ion ($[H^+]$) concentration and is given by the formula: $\text{pH} = -\log[H^+]$

A decrease in pH means an increase in the $[H^+]$ concentration.

Given Data

- Initial pH = 6
- Dilution factor = 100 times

Concept

- pH formula: $\text{pH} = -\log[H^+]$
- Effect of dilution on pH: When a solution is diluted 100 times, the concentration of H^+ ions decreases by a factor of 100.

v Contribution of water auto-ionization:

Water itself contributes $[H^+] = 10^{-7} \text{ M}$, because $K_w = [H^+][OH^-] = 10^{-14}$

Calculation1. Find the initial $[H^+]$ concentration: $[H^+] = 10^{-6} \text{ M}$

2. New concentration after dilution (100 times):

$$[H^+] = 10^{-6}/100 = 10^{-8} \text{ M}$$

3. Total $[H^+]$:

$$[H^+]_{\text{total}} = 10^{-8} + 10^{-7} = 1.1 \times 10^{-7} \text{ M}$$

4. **New pH:** $\text{pH} = -\log(1.1 \times 10^{-7}) = 7 - \log(1.1) = 6.98$ **Answer** option 1, (6.98)70. **Ans (3)**

Telegram: @NEETxNOAH

Explanation

An amphiprotic species is a molecule or ion that can both donate and accept a proton (H^+). These species can act as both Bronsted-Lowry acids and Bronsted-Lowry bases.

Given Data

The species given in the question:

1. $H_2PO_2^-$ (Hypophosphite ion)
2. HPO_3^{2-} (Metaphosphate ion)
3. HCO_3^- (Bicarbonate ion)
4. $CH_3CO_2^-$ (Acetate ion)
5. HPO_4^{2-} (Hydrogen phosphate ion)

Concept

To be amphiprotic, a species must:

- Have at least one proton (H^+) to donate (acidic behavior).
- Have the ability to accept a proton (basic behavior).

Now, we check each species:

 $H_2PO_2^- \rightarrow$ Cannot accept a proton easily,

1. only acts as an acid (not amphiprotic).

 $HPO_3^{2-} \rightarrow$ Cannot donate a proton easily,

2. only acts as a base (not amphiprotic).

 HCO_3^- (Bicarbonate) $\rightarrow$ Can donate H^+ to3. form CO_3^{2-} and accept H^+ to form H_2CO_3 . $CH_3CO_2^-$ (Acetate) $\rightarrow$ Only accepts a

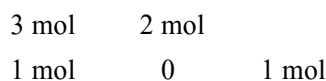
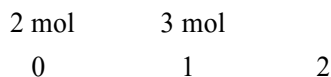
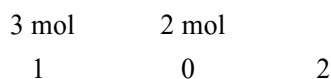
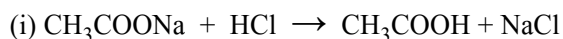
4. proton, does not donate.

 HPO_4^{2-} (Hydrogen phosphate) $\rightarrow$ Can donate H^+ to form PO_4^{3-} and accept H^+ to5. form $H_2PO_4^-$.

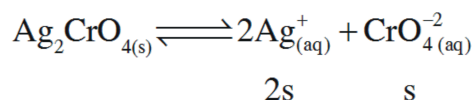
Answer option 3, (III), (V)

74. **Ans (2)**NCERT XIth Part - 1 Pg. # 220

75. Ans (2)



78. Ans (2)



$$4s^3 = K_{sp}$$

$$4s^3 = y$$

$$s = \left(\frac{y}{4}\right)^{\frac{1}{3}}$$

$$[\text{Ag}^+] = 2s = 2 \times \left(\frac{y}{4}\right)^{\frac{1}{3}}$$

$$= \left(\frac{8 \times y}{4}\right)^{\frac{1}{3}} = (2y)^{\frac{1}{3}}$$

80. Ans (2)

Metallic iron is oxidised to Fe^{3+}

89. Ans (2)

$$E = \frac{M_W}{V.f.}; E = \frac{(M_W)\text{SO}_2}{4} = \frac{64}{4} = 16$$

90. Ans (2)

Telegram: @NEETxNOAH

(1) xyz

$$+3 - 5 + 1 \neq 0$$

(2) xyz₂

$$+3 - 5 + 1 \times 2 = 0$$

(3) x₂yz

$$3 \times 2 - 5 + 1 \neq 0$$

(4) (xy)₂z

$$(+3 - 5) \times 2 + 1 \neq 0$$

91. Ans (2)

NCERT Pg. 51

93. Ans (1)

Conduction of gametes is function of secondary sex organs.

94. Ans (2)

NCERT Pg. 50

99. Ans (4)

NCERT Pg. # 51

100. Ans (3)

NCERT-XII, Pg. # 33

101. Ans (1)

NCERT-XII, Pg. # 36, 37

104. Ans (4)

NCERT Pg. # 51

105. Ans (2)

NCERT-XIIth Pg. 33

108. Ans (4)

NCERT Pg. # 44

109. Ans (2)

NCERT Pg. # 47

112. Ans (1)

NCERT Pg. 54

113. Ans (2)

NCERT Pg # 46

115. Ans (4)

NCERT-XII, Pg. # 27

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|--|--|
| <p>119. Ans (1)
NCERT-XII Pg#37</p> <p>121. Ans (4)
NCERT Pg. # 50(E), 55(H)</p> <p>122. Ans (1)
Module-1 Pg. # 51</p> <p>123. Ans (3)
NCERT-XII, Pg. # 34, Fig-2.9</p> <p>124. Ans (4)
NCERT XII Pg # 46</p> <p>128. Ans (4)
NCERT XII - Page no 43</p> <p>129. Ans (3)
NCERT Pg. # 32</p> <p>130. Ans (4)
NCERT-XII, Pg. # 34,35</p> <p>131. Ans (2)
NCERT Pg. # 43/44</p> <p>133. Ans (4)
NCERT Pg. # 48(E), 53(H)</p> <p>135. Ans (2)
XII NCERT Pg. 54, Para - 3.6</p> <p>137. Ans (1)
NCERT (XII) Pg. # 15</p> <p>138. Ans (3)
NCERT XII, Page # 20, 21</p> <p>141. Ans (2)
XII NCERT Pg # 26</p> <p>142. Ans (3)
NCERT-XIIth, Page 26 (E), 27 (H)</p> <p>143. Ans (2)
In majority of angiosperm female gametophyte is 7 celled 8 nucleated structure</p> | <p>144. Ans (3) Telegram: @NEETxNOAH
NCERT XII Page # 28</p> <p>151. Ans (1)
NCERT-XII, Pg. # 12</p> <p>152. Ans (3)
NCERT XII Page. No. 35 (E) 37 (H)</p> <p>153. Ans (2)
NCERT XII Page. No. 36 (E) 38 (H)</p> <p>156. Ans (1)
NCERT XII Pg # 29</p> <p>157. Ans (2)
NCERT XIIth Pg.# 26,27</p> <p>158. Ans (3)
NCERT (XIIth) Pg. # 28</p> <p>166. Ans (3)
NCERT XII page 24</p> <p>167. Ans (3)
NCERT XII page 34, 35</p> <p>168. Ans (4)
NCERT XII Pg # 30, fig. 2.1 (a), 34, 36, 38, 39
Nucellar adventive embryony is found in Citrus and mango.</p> <p>169. Ans (1)
NCERT-XI Pg. # 27</p> <p>170. Ans (1)
NCERT-XII-Page No. # 38, 39</p> <p>172. Ans (4)
NCERT XII Pg. # 34(E), 36(H)</p> <p>179. Ans (1)
NCERT Pg. # 38(E), 41(H)</p> <p>180. Ans (1)
NCERT Pg # 38</p> |
|--|--|